

LAB COURSE OUTLINE

01.	Faculty	Faculty of Engineering & Applied Science (FEAS)			
02.	Department	Computer Science and Engineering (CSE)			
03.	Program	B.Sc. in CSE			
04.	Name of Course	Physics I Lab			
05.	Course Code	PHY 102			
06.	Bi- semester	Spring 2024			
07.	Pre-requisites	N/A			
08.	Co-requisite	Physics I (PHY 101)			
08.	Course Type	Core Course			
09.	Credit Hours	1.5			
10.	Intake/ Section	53/S1 (Day)			
11.	Class Schedule	Intake & Section	Class Day	Class Hours	Venue
		53/S1 (Day)	Tuesday	4:00 PM – 6:30 PM	Room No.: B-3/R-902
12.	Campus Location	Permanent Campus			
13.	Course Website				
14.	Course Instructor Detail	Name: Md. Sharif Ahmed		Designation: Lecturer	
		Specialization: Condensed Matter Physics, Theoretical Physics, Computational Engineering, Materials Engineering, Solid State, Nuclear Physics Physics etc.			
		Room No. B-1/602		Email: sharifahmed@bubt.edu.bd	Cell No. 01567828688
15.	Counseling Hours	Intake & Section	Day		Counseling Hours
		53/S1	Monday		1:30 PM – 3:00 PM
16.	Text Book	[1] Practical Physics, Dr. Giasuddin Ahmad and Md. Shahabuddin			
17.	Reference	[1] Physics I Lab Manual Sheets, CSE Department, BUBT [2] Practical Physics, R.K. Shukla and Anchal Srivastava			
18.	Equipment & Aids	Students must carry learning materials (<i>calculator, notebook, pen, etc.</i>) in classroom. Borrowing learning materials in exam or classroom from fellow students is strictly prohibited. <i>Students are also advised to keep at least one sheet of blank paper (A4 size) with them every class.</i>			
19.	Course Rationale	This course will provide practical knowledge by applying the experimental methods to correlate with the Physics theory and the ideas of different measuring devices and meters to record the data with precision.			
20.	Course Synopsys	This course will mainly focus on Waves and Oscillations: Compound pendulum, Katter’s pendulum, Spring- Mass system Heat & Thermodynamics: Thermal conductivity, Pressure coefficient Properties of Matter: Modulus of rigidity, Young’s Modulus, Surface Tension Physical Optics: Refractive Index			
21.	Course Outcomes (COs)	Upon completion of this course students will be able to: CO1: To apply the basic laws and theories of classical physics to characterize physical properties of various states of materials and mediums CO2: Design and develop the experimental setup to evaluate simple pendulum, etc. CO3: Select and use modern hardware and software tools and devices CO4: Assess societal, health, safety, legal and cultural issues involved with the mini project			

	CO5: Realize the impact of societal, environmental and sustainable development issues for the design solution of mini project. CO6: Apply professional ethics and responsibilities in the implementation of mini project. CO7: Work individually and in a team CO8: Communicate and share knowledge, data, information, results etc. with others CO9: Apply engineering project management knowledge and skill to implement the mini project. CO10: Gather and apply knowledge, data and information from various multidisciplinary sources to analyze, design and implement the mini project																																																																																																																				
	<table><tr><th colspan="10">CO Attainment Process</th></tr><tr><th rowspan="2">CO No.</th><th rowspan="2">PO No.</th><th colspan="3">Bloom's Domain</th><th rowspan="2">WK</th><th rowspan="2">WP</th><th rowspan="2">EA</th><th rowspan="2">Delivery Methods / Activities</th><th colspan="2" rowspan="2">Assessment Tools</th></tr><tr><th>C</th><th>A</th><th>P</th></tr><tr><td>CO1</td><td>PO1</td><td>5</td><td></td><td></td><td>K3</td><td>WP1</td><td></td><td rowspan="10"> • Class Lecture • Lab Experiment • Tutorial • Group Discussion • Lab Practice </td><td colspan="2" rowspan="10"> • Mini Project Presentation • Attendance • Continuous Lab Performance • Final lab Test/Exam • Lab Report • Individual/Team Work Rubrics • Presentation, Viva and Lab Report </td></tr><tr><td>CO2</td><td>PO4</td><td>5</td><td></td><td></td><td>K3</td><td>WP1</td><td></td></tr><tr><td>CO3</td><td>PO5</td><td></td><td>4</td><td>5</td><td>K5</td><td>WP1</td><td></td></tr><tr><td>CO4</td><td>PO6</td><td></td><td></td><td>5</td><td>K6</td><td></td><td></td></tr><tr><td>CO5</td><td>PO7</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>CO6</td><td>PO8</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>CO7</td><td>PO9</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>CO8</td><td>PO10</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>CO9</td><td>PO11</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>CO10</td><td>PO12</td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>										CO Attainment Process										CO No.	PO No.	Bloom's Domain			WK	WP	EA	Delivery Methods / Activities	Assessment Tools		C	A	P	CO1	PO1	5			K3	WP1		• Class Lecture • Lab Experiment • Tutorial • Group Discussion • Lab Practice	• Mini Project Presentation • Attendance • Continuous Lab Performance • Final lab Test/Exam • Lab Report • Individual/Team Work Rubrics • Presentation, Viva and Lab Report		CO2	PO4	5			K3	WP1		CO3	PO5		4	5	K5	WP1		CO4	PO6			5	K6			CO5	PO7							CO6	PO8							CO7	PO9							CO8	PO10							CO9	PO11							CO10	PO12						
CO Attainment Process																																																																																																																					
CO No.	PO No.	Bloom's Domain			WK	WP	EA	Delivery Methods / Activities	Assessment Tools																																																																																																												
		C	A	P																																																																																																																	
CO1	PO1	5			K3	WP1		• Class Lecture • Lab Experiment • Tutorial • Group Discussion • Lab Practice	• Mini Project Presentation • Attendance • Continuous Lab Performance • Final lab Test/Exam • Lab Report • Individual/Team Work Rubrics • Presentation, Viva and Lab Report																																																																																																												
CO2	PO4	5			K3	WP1																																																																																																															
CO3	PO5		4	5	K5	WP1																																																																																																															
CO4	PO6			5	K6																																																																																																																
CO5	PO7																																																																																																																				
CO6	PO8																																																																																																																				
CO7	PO9																																																																																																																				
CO8	PO10																																																																																																																				
CO9	PO11																																																																																																																				
CO10	PO12																																																																																																																				
22.	Teaching Methods	Lecture, Laboratory hardware and software experiments, Project developments.																																																																																																																			
23.	Topic Outline																																																																																																																				
	Expt. No	Selected Topics				COs	Reading Reference	Activities																																																																																																													
	1	Familiarization with Vernier Calipers and Screw Gauge by measuring various samples.				CO1, CO3, CO8	• Text [1] • Lab sheet 01	• Class Lecture • Lab Experiment • Tutorial • Group Discussion • Lab Practice • Mini Project • Presentation																																																																																																													
	2	To determine the value of g, acceleration due to gravity, by means of a compound pendulum.				CO1, CO3, CO8	• Text [1] • Lab Sheet 02																																																																																																														
	3	To determine the spring constant and effective mass of a given spiral spring.				CO1, CO3, CO8	• Text [1] • Lab sheet 03																																																																																																														
	4	To determine Young's modulus by the flexure of a beam.				CO1, CO3, CO8	• Text [1] • Lab Sheet 04																																																																																																														
	5	To determine the surface tension of water by capillary tube method.				CO1, CO3, CO8	• Text [1] • Lab Sheet 05																																																																																																														
	6	To determine the modulus of rigidity of a wire by Statistical method.				CO1, CO3, CO8	• Text [1] • Lab Sheet 06																																																																																																														
	7	To determine the modulus of rigidity of a wire by method of oscillations (dynamic method).				CO1, CO3, CO8	• Text [1] • Lab Sheet 07																																																																																																														
	8	To determine the pressure co-efficient of a gas at constant volume by constant volume air thermometer.				CO1, CO3, CO8	• Text [1] • Lab Sheet 08																																																																																																														
	9	To determine the refractive index of the material of a prism.				CO1, CO3, CO8	• Text [1] • Lab Sheet 09																																																																																																														

	10	To determine the thermal conductivity of a bad conductor by Lees and Chorlton’s method.	CO1, CO3, CO8	• Text [1] • Lab Sheet 10																					
	11	To verify the Newton’s law of cooling of different materials and different liquids	CO1, CO3, CO8	• Text [1] • Lab Sheet 11																					
	12	To determine the value of g, acceleration due to gravity, by means of Kater’s Pendulum	CO1, CO3, CO8	• Text [1] • Lab Sheet 12																					
		Final Examination																							
24.	Assessment and Marks Distribution:	Students will be assessed on the basis of their overall performance in all the exams, quizzes, and class participation. Final grade will be keyed based on: ❖ Attendance (5%) ❖ Lab Report (10%) ❖ Mini Project, Presentation and Viva (25%) ❖ Continuous Lab Performance (CLP) (30%) ❖ Lab Final (30%)																							
25.	Assessment Methods of COs	Assessment methods of COs are given below:																							
		Assessment																							
		COs	Attendance	Lab Report	Mini Project show and Presentation	CLP	Lab Final Test/Open ended Lab Test																		
		CO1		√	√		√																		
		CO2			√	√	√																		
		CO3		√	√	√	√																		
		CO4			√																				
		CO5			√																				
		CO6			√																				
		CO7			√	√																			
		CO8		√	√	√																			
		CO9			√																				
		CO10			√																				
26.	Grading Policy	The following chart will be followed for grading. This has been customized from the guideline provided by the School of Engineering and Computer Science.																							
		<table><tr><td>A+</td><td>A</td><td>A-</td><td>B+</td><td>B</td><td>B-</td><td>C+</td><td>C</td><td>D</td><td>F</td></tr><tr><td>≥ 80</td><td>75-<80</td><td>70-<75</td><td>65-<70</td><td>60-<65</td><td>55-<60</td><td>50-<55</td><td>45-<50</td><td>40-<45</td><td><40</td></tr></table>				A+	A	A-	B+	B	B-	C+	C	D	F	≥ 80	75-<80	70-<75	65-<70	60-<65	55-<60	50-<55	45-<50	40-<45	<40
A+	A	A-	B+	B	B-	C+	C	D	F																
≥ 80	75-<80	70-<75	65-<70	60-<65	55-<60	50-<55	45-<50	40-<45	<40																
27.	Additional Course Policies	Lab Reports:	Report on previous Experiment must be submitted before the beginning of new experiment. A bonus may be obtained if a student submits a neat, clean and complete lab report.																						
		Presentation :	There will be one presentation for each student. Presentation will be taken individually or group work. Capstone project will be given and student must have to present their project.																						
		Exams :	Mid-term and final exam will be closed book, closed notes. Mobile is strictly prohibited in exam hall. Please bring your own watch and synchronize time during exam hours.																						
		Test Policy :	If you are absent from a test, and you have not spoken to the teacher personally beforehand, your grade for the test will be zero. No make-up for class test will be taken because it has alternative (two out of three).																						
28.	Rubrics for Mini Project Presentation, Report writing	A project should be presented on any topic related to CO3 in a group of two/three focusing on the course outcome, CO3 . Finally, a project report based on the presented project must be submitted to receive the 25% marks of Mini Project, Presentation and Viva. A project presentation content must contain the following: 1. Title Page (Project title, Student(s) name and ID with department name & BUBT logo) 2. Background, Motivation and Objectives 3. Literature Review																							

	4. Methodology (Block diagram) 5. Detailed Circuit Diagram 6. Simulation / Hardware Prototype 7. Results and Discussions 8. Conclusion A project report must be submitted according to the following format: 1. Project Title, Student(s) name and ID with department name & BUBT logo 2. Introduction 3. Background, Motivation and Objectives 4. Literature Review 5. Methodology 6. Proposed method 7. Simulation / Hardware Prototype 8. Results and Discussions 9. Socio-economic Impact 10. Environmental impact & Sustainability 11. Conclusion 12. References
--	--

29 Bloom's Taxonomy for Teaching-Learning

Bloom's Taxonomy is a set of three hierarchical models used to classify educational learning objectives into levels of complexity and specificity. The three lists cover the learning objectives in Cognitive, Affective and Psychomotor domains. The Cognitive domain list has been the primary focus of most education and is frequently used to structure curriculum learning objectives, assessments and activities. The three domains and respective levels are illustrated below.

Cognitive [C] (Knowledge-based)	Affective [A] (Emotion-based)	Psychomotor [P] (Action-based)
1. Remembering	1. Receiving	1. Imitating
2. Understanding	2. Responding	2. Manipulating
3. Applying	3. Valuing	3. Précising
4. Analyzing	4. Organizing	4. Articulating
5. Evaluating	5. Characterizing	5. Naturalizing
6. Creating	--- --- ---	--- --- ---

Graduate Attributes for B.Sc. in Engineering Program based on Washington Accord

30 Program Outcomes (POs)

POs are narrower statements that describe what students are expected to know and be able to do by the Time of graduation. These relate to the knowledge skills and attitudes that students acquire while progressing through the program. The students of the B.Sc. in EEE program are expected to achieve the following graduate attributes or program outcomes at the time of graduation.

PO1–Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.

PO2–Problem analysis: Identify, formulate, research the literature and analyze complex engineering problems and reach substantiated conclusions using first principles of mathematics, the natural sciences and the engineering sciences.

PO3–Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety as well as cultural, societal and environmental concerns.

PO4–Investigation: Conduct investigations of complex problems, considering design of experiments, analysis and interpretation of data and synthesis of information to provide valid conclusions.

PO5–Modern tool usage: Create, select and apply appropriate techniques, resources and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6–The engineer and society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.

PO7–Environment and sustainability: Understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of, and need for sustainable development.

PO8–Ethics: Apply ethical principles and commit to professional ethics, responsibilities and the norms of the engineering practice.

	PO9–Individual work and teamwork: Function effectively as an individual and as a member or leader of diverse teams as well as in multidisciplinary settings. PO10–Communication: Communicate effectively about complex engineering activities with the engineering community and with society at large. Be able to comprehend and write effective reports, design documentation, make effective presentations and give and receive clear instructions. PO11–Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one’s own work as a member or a leader of a team to manage projects in multidisciplinary environments. PO12–Life-long learning: Recognize the need for and have the preparation and ability to engage in independent, life-long learning in the broadest context of technological change.
31	Knowledge Profile (WK) WK1: A systematic, theory-based understanding of the natural sciences applicable to the discipline. WK2: Conceptually-based mathematics, numerical analysis, statistics and formal aspects of computer and information science to support analysis and modeling applicable to the discipline. WK3: A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline. WK4: Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline. WK5: Knowledge that supports engineering design in a practice area. WK6: Knowledge of engineering practice (technology) in the practice areas in the engineering discipline. WK7: Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline: ethics and the professional responsibility of an engineer to public safety; the impacts of engineering activity: economic, social, cultural, environmental and sustainability. WK8: Engagement with selected knowledge in the research literature of the discipline.
32	Range of Complex Engineering Problem Solving Complex Engineering Problems have characteristic WP1 and some or all of WP2 to WP7: WP1–Depth of knowledge required: Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6 or K8 which allows a fundamentals-based, first principles analytical approach WP2–Range of conflicting requirements: Involve wide-ranging or conflicting technical, engineering and other issues WP3–Depth of analysis required: Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models WP4–Familiarity of issues: Involve infrequently encountered issues WP5–Extent of applicable codes: Are outside problems encompassed by standards and codes of practice for professional engineering WP6–Extent of stakeholder involvement and conflicting requirements: Involve diverse groups of stakeholders with widely varying needs WP7–Interdependence: Are high level problems including many component parts or sub-problems
33	Range of Complex Engineering Activities Attribute Complex activities means (engineering) activities or projects that have some or all of the following characteristics: EA1–Range of resources: Involve the use of diverse resources (and for this purpose resources include people, money, equipment, materials, information and technologies) EA2–Level of interaction: Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering or other issues EA3–Innovation: Involve creative use of engineering principles and research based knowledge in novel ways EA4–Consequences for society and the environment: Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation EA5–Familiarity: Can extend beyond previous experiences by applying principles-based approaches.
34	Our promises are based on the three cardinal principles: (a) What we do believe (b) What we do practice, and (c) What we will promote However, students are advised to undertake the following commitments for moral development.

1. To be punctual and attentive in class 2. To maintain inclusive learning environment 3. To ensure mutual respect 4. To be cooperative in group learning. 5. To be innovative and Creative 6. To follow dress code and wearing ID card 7. To be always proactive	8. Try to follow and review day to day class 9. To avoid conspiracy 10. To prioritize honesty & faith 11. To be motivated for asking question and encourage feedback 12. To develop attitude for speaking in English 13. Do not ignore to carry out any assignments or commitments 14. To be clean and decent in all levels.	15. To be sincere for class preparation 16. Do not forget to switch-off the cellphone in class 17. Do not forget to carry course pack and learning stuffs in class 18. To maintain loyalty and trust to the university 19. Must avoid unfair means and plagiarism in exam, reports and assignments 20. Must maintain eco-friendly environment in the campus.
Social & Moral Capital		